Visualization Activities with Datasets

Python Code, Output Visualizations and Answers

Prepared as a complete activity record using Python, Pandas and Matplotlib.

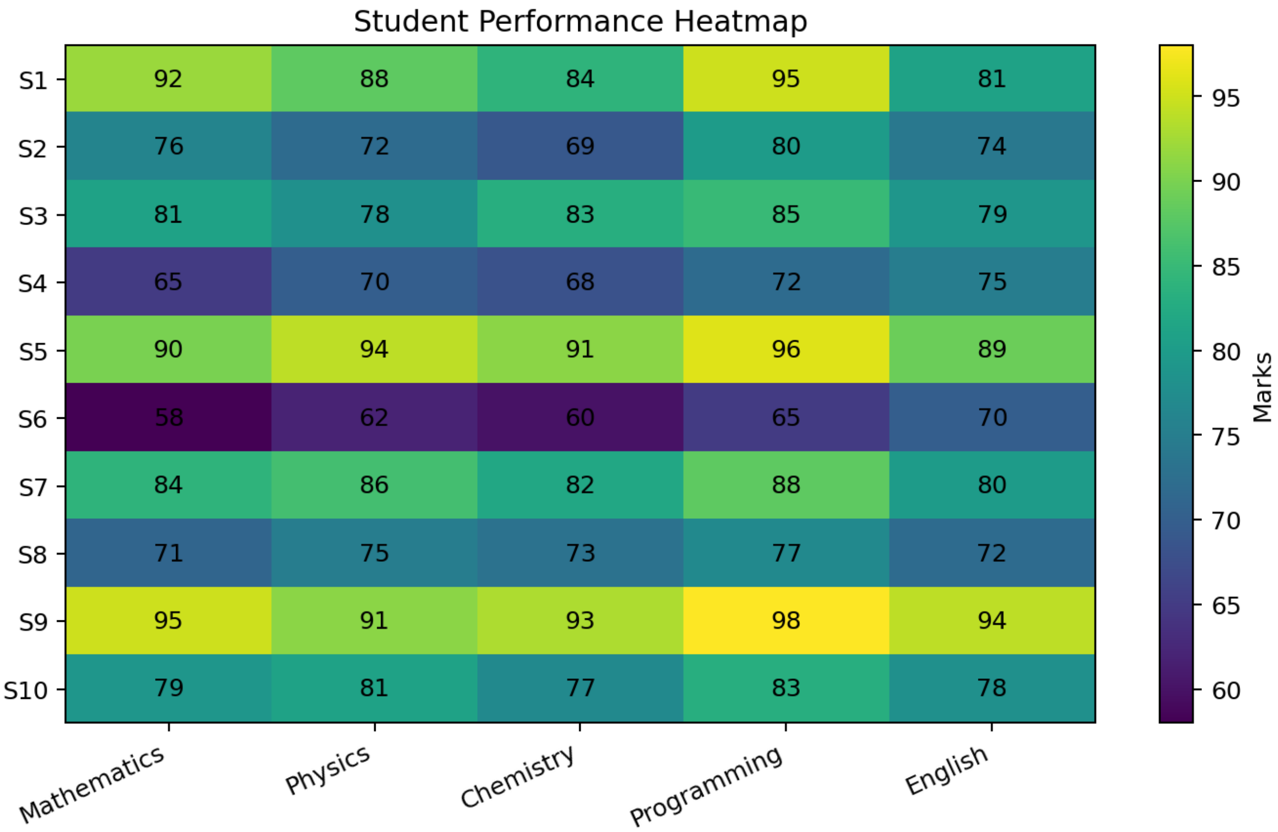
Activity 1: Heatmap - Student Performance Analysis

Student	Mathematics	Physics	Chemistry	Programming	English
S1	92	88	84	95	81
S2	76	72	69	80	74
S3	81	78	83	85	79
S4	65	70	68	72	75
S5	90	94	91	96	89
S6	58	62	60	65	70
S7	84	86	82	88	80
S8	71	75	73	77	72
S9	95	91	93	98	94
S10	79	81	77	83	78

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame(data, index=students, columns=subjects)
plt.imshow(df.values, aspect="auto")
plt.xticks(range(len(df.columns)), df.columns, rotation=25)
plt.yticks(range(len(df.index)), df.index)
for i in range(df.shape[0]):
    for j in range(df.shape[1]):
        plt.text(j, i, df.iloc[i,j], ha="center", va="center")
plt.colorbar(label="Marks")
plt.title("Student Performance Heatmap")
plt.tight_layout()
plt.show()
```

Output



Questions and Answers

1. Construct a Heatmap.

   Answer: The heatmap is shown above. Cell intensity visually compares marks across students and subjects.

2. Identify highest overall performer.

   Answer: S9 is the highest overall performer with an average of **94.2** marks.

3. Find highest average subject.

   Answer: Programming has the highest subject average of **83.9**.

4. Identify students needing support.

   Answer: Using an overall average below 70 as the support criterion, **S6** need support. S4 is also comparatively low and may be monitored.

5. Explain Heatmap advantages.

   Answer: A heatmap quickly reveals high and low values, patterns, subject-wise strengths, weak areas and outliers using colour intensity.

Activity 2: Violin Plot & Boxplot - Employee Salary Analysis

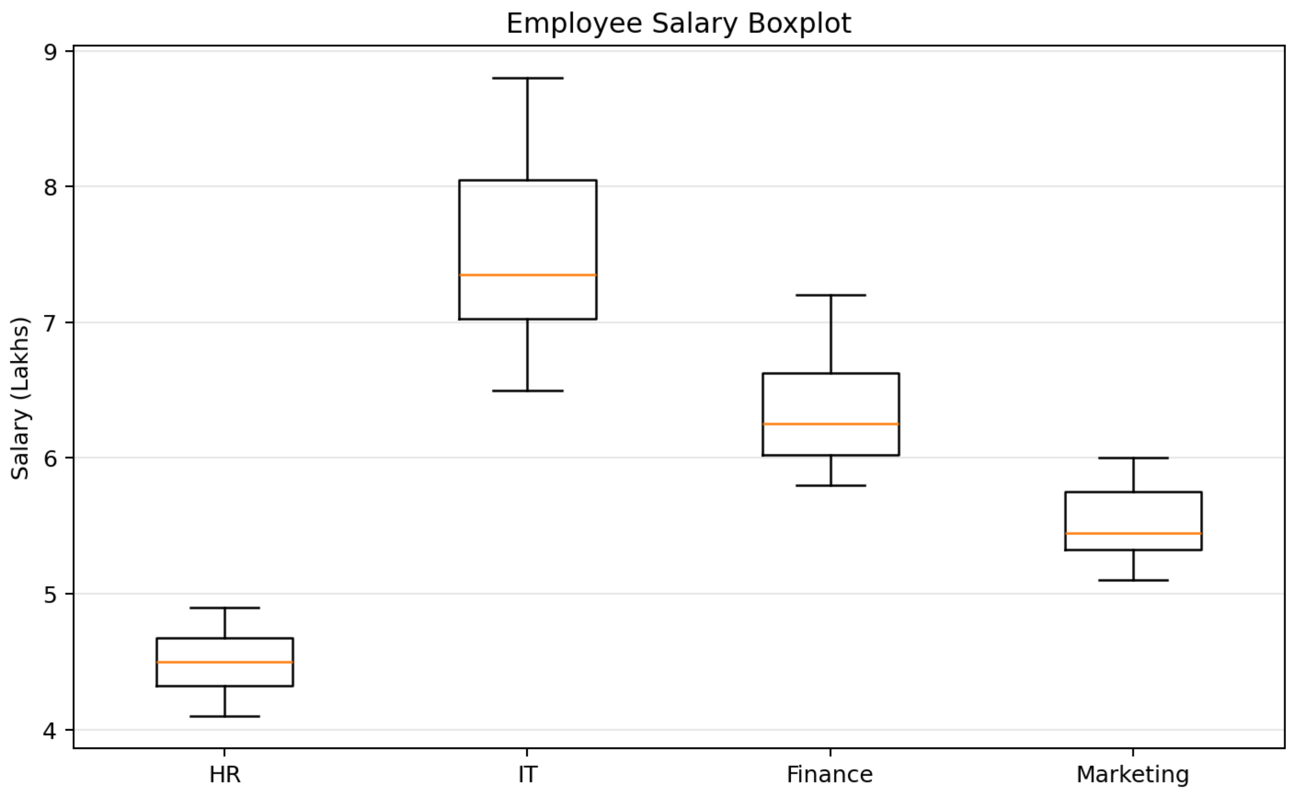
Employee	HR	IT	Finance	Marketing
E1	4.2	6.5	5.8	5.1
E2	4.5	7.0	6.2	5.3
E3	4.7	7.4	6.0	5.5
E4	4.1	6.8	5.9	5.4
E5	4.6	8.2	6.7	5.8
E6	4.3	7.6	6.4	5.6
E7	4.8	8.5	6.9	5.9
E8	4.4	7.1	6.1	5.2
E9	4.9	8.8	7.2	6.0
E10	4.5	7.3	6.3	5.4

Python Code

```
import matplotlib.pyplot as plt

plt.violinplot([df[c] for c in df.columns],
               showmeans=True, showmedians=True)
plt.xticks(range(1, 5), df.columns)
plt.ylabel("Salary (Lakhs)")
plt.title("Employee Salary Violin Plot")
plt.show()

plt.boxplot([df[c] for c in df.columns],
            tick_labels=df.columns)
plt.ylabel("Salary (Lakhs)")
plt.title("Employee Salary Boxplot")
plt.show()
```

Output 1: Violin Plot

Output 2: Boxplot



Questions and Answers

1. Draw violin plots.

   Answer: The violin plot above displays the salary distribution for HR, IT, Finance and Marketing.

2. Construct boxplots.

   Answer: The boxplot above summarizes median, quartiles and spread for each department.

3. Largest variation?

   Answer: **IT** has the largest range (2.3 lakhs), showing the greatest salary variation.

4. Highest median?

   Answer: **IT** has the highest median salary of **7.35 lakhs**.

5. Which plot better represents distribution?

   Answer: The **violin plot** better represents the full distribution because it shows density and distribution shape. The boxplot is better for a compact median, quartile and outlier summary.

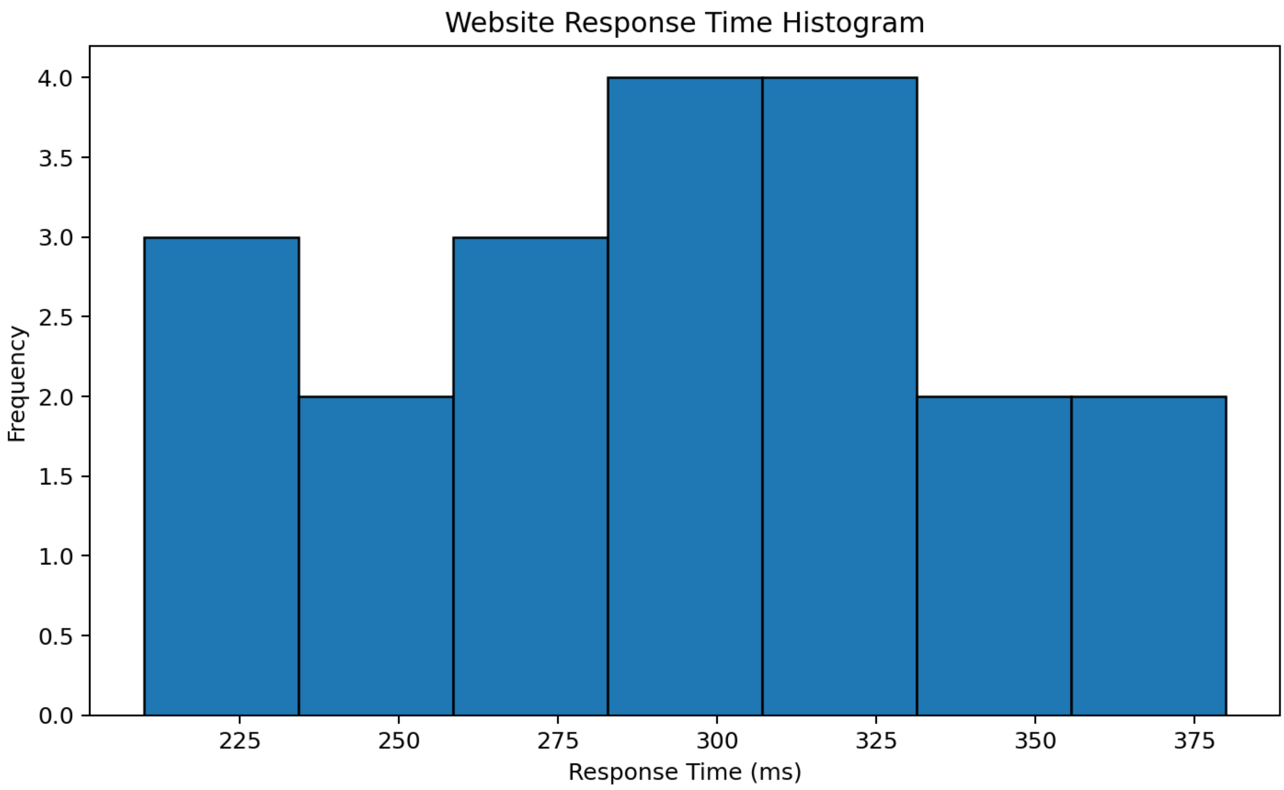
Activity 3: Histogram & Density Plot - Website Response Time

Obs	Response Time (ms)
1	210
2	225
3	230
4	240
5	250
6	260
7	270
8	280
9	285
10	290
11	300
12	305
13	310
14	315
15	320
16	330
17	340
18	350
19	360
20	380

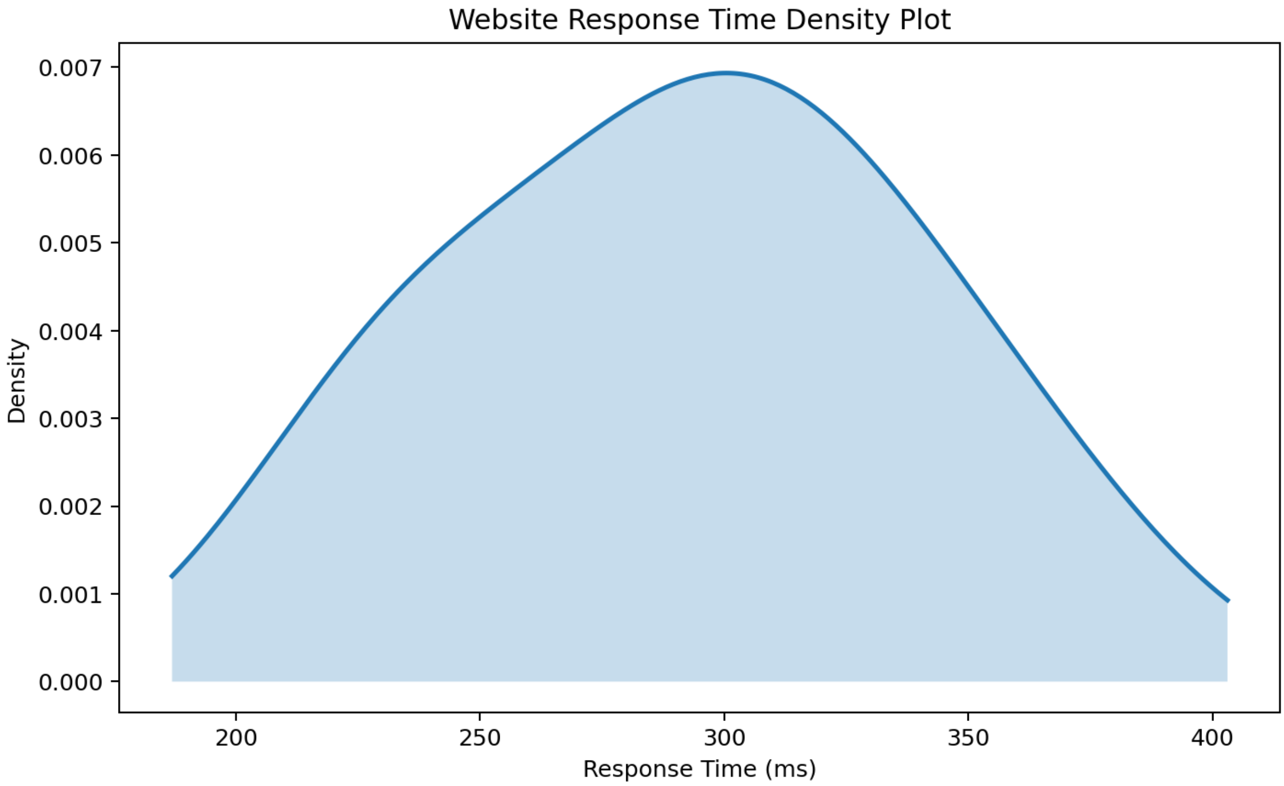
Python Code

```
import matplotlib.pyplot as plt
import pandas as pd

response = [210,225,230,240,250,260,270,280,285,290,
            300,305,310,315,320,330,340,350,360,380]

plt.hist(response, bins=7, edgecolor="black")
plt.xlabel("Response Time (ms)")
plt.ylabel("Frequency")
plt.title("Website Response Time Histogram")
plt.show()

pd.Series(response).plot(kind="density")
plt.xlabel("Response Time (ms)")
plt.title("Website Response Time Density Plot")
plt.show()
```

Output 1: Histogram



Output 2: Density Plot



Questions and Answers

1. Draw histogram.

   Answer: The histogram above groups website response times into frequency intervals.

2. Plot density.

Answer: The density plot above provides a smooth estimate of the response-time distribution.

3. Determine skewness.

   Answer: The sample skewness is approximately **-0.008**. This indicates an approximately symmetric distribution with a very slight left skew.

4. Common range.

   Answer: Most observations are concentrated roughly between **250 ms and 340 ms**.

5. Which plot better represents distribution?

   Answer: The **density plot** better shows the smooth overall distribution shape, while the histogram is more direct for actual frequency ranges.

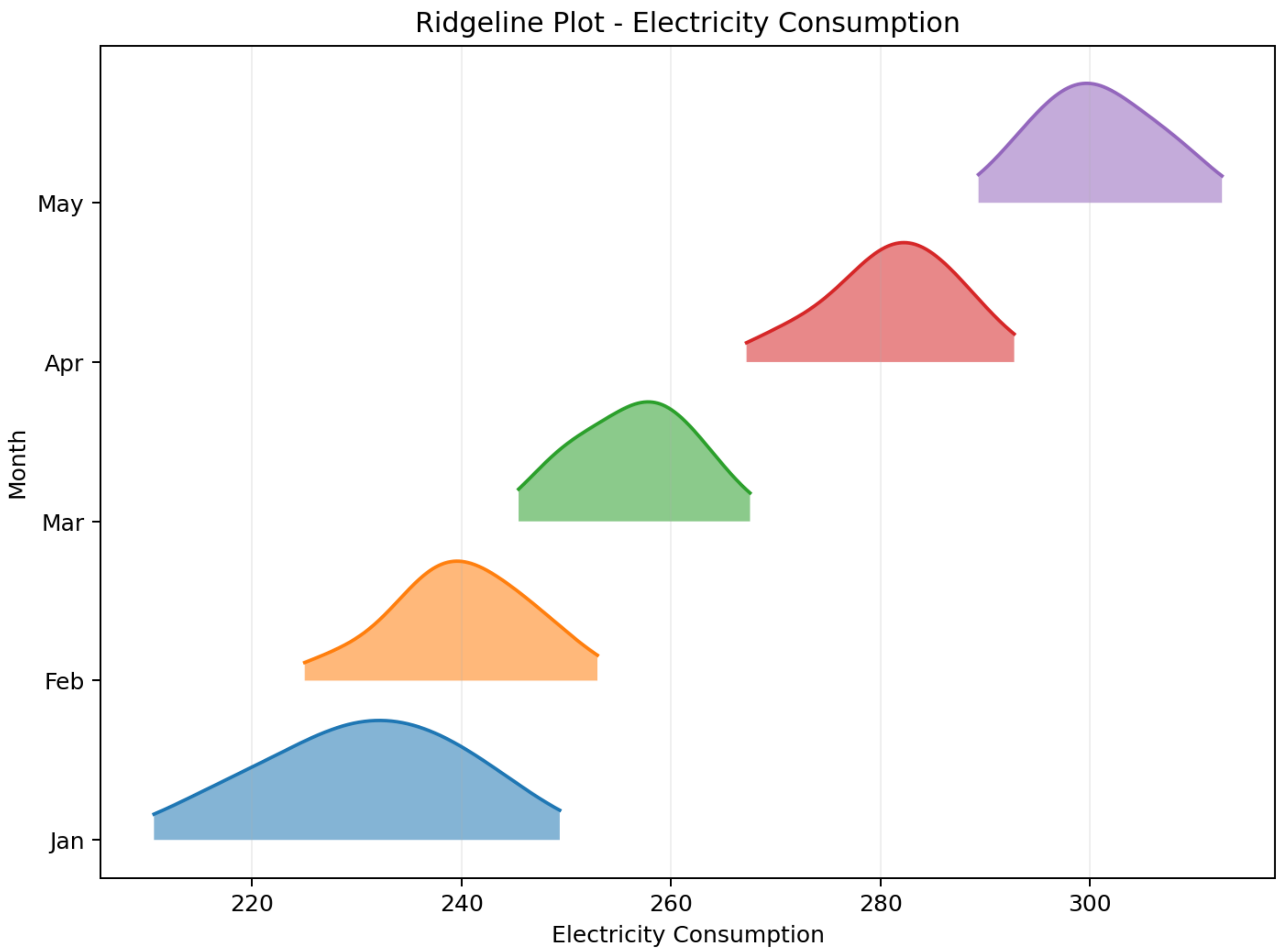
Activity 4: Ridgeline Plot - Electricity Consumption

House	Jan	Feb	Mar	Apr	May
H1	220	235	250	275	295
H2	240	245	260	285	305
H3	230	238	255	280	300
H4	215	228	248	270	292
H5	225	240	258	282	298
H6	245	250	265	290	310
H7	235	242	259	284	302
H8	228	236	252	278	297
H9	238	246	262	287	307
H10	232	239	256	281	301

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

months = df.columns
for i, month in enumerate(months):
    values = df[month].values
    # KDE values x and y are calculated for each month
    x, y = kde(values)
    y = y / y.max() * 0.75
    plt.fill_between(x, i, i+y, alpha=0.55)
    plt.plot(x, i+y)

plt.yticks(range(len(months)), months)
plt.xlabel("Electricity Consumption")
plt.ylabel("Month")
plt.title("Ridgeline Plot - Electricity Consumption")
plt.show()
```

Output



Questions and Answers

1. Create ridgeline plot.

   Answer: The ridgeline plot above compares monthly electricity-consumption distributions.

2. Highest average month?

   Answer: May has the highest average consumption of **300.7**.

3. Widest spread?

   Answer: Jan has the widest range of **30** units.

4. Describe trend.

   Answer: Electricity consumption shows a clear increasing trend from January to May. The distribution shifts steadily toward higher consumption each month.

5. Why ridgeline?

   Answer: A ridgeline plot allows several distributions to be compared vertically, making shifts in location, shape and spread across months easy to observe.

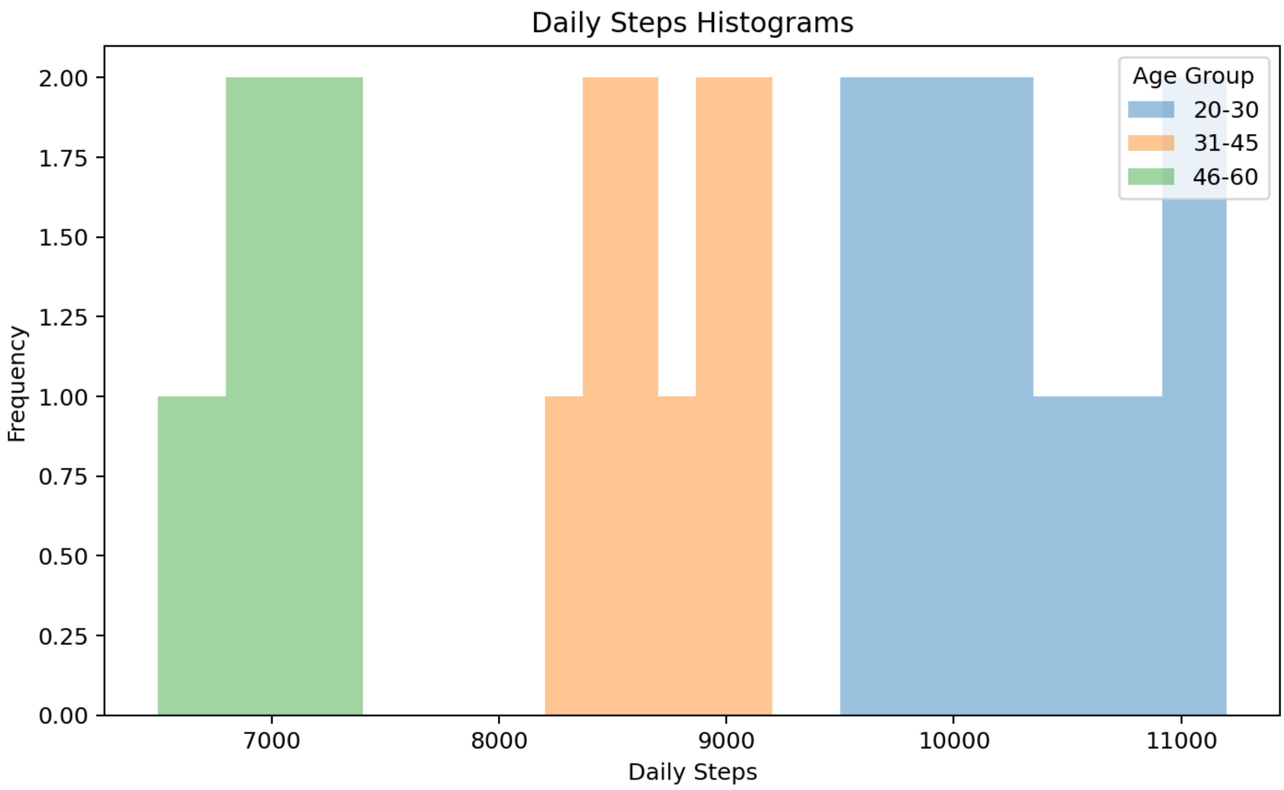
Activity 5: Histogram, Density & Boxplot - Daily Steps

Person	20-30	31-45	46-60
P1	9500	8200	6500
P2	10200	8600	6700
P3	9800	8400	6900
P4	11000	9000	7200
P5	10500	8900	7000
P6	9700	8500	6800
P7	10100	8700	7100
P8	10800	9100	7300
P9	11200	9200	7400
P10	9900	8600	6950

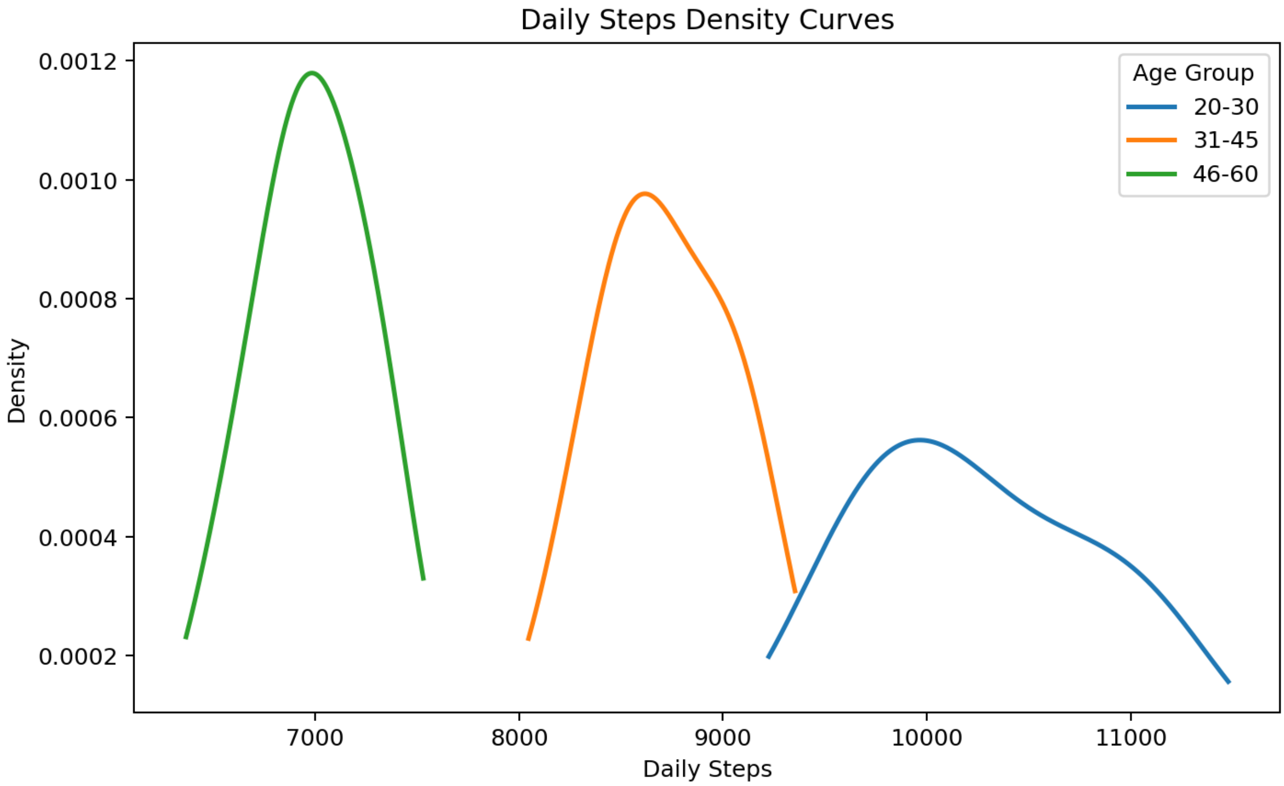
Python Code

```
import matplotlib.pyplot as plt

for c in df.columns:
    plt.hist(df[c], bins=6, alpha=0.45, label=c)
plt.xlabel("Daily Steps")
plt.ylabel("Frequency")
plt.legend(title="Age Group")
plt.title("Daily Steps Histograms")
plt.show()

for c in df.columns:
    df[c].plot(kind="density", label=c)
plt.xlabel("Daily Steps")
plt.legend(title="Age Group")
plt.title("Daily Steps Density Curves")
plt.show()

plt.boxplot([df[c] for c in df.columns],
            tick_labels=df.columns)
plt.xlabel("Age Group")
plt.ylabel("Daily Steps")
plt.title("Daily Steps Boxplots")
plt.show()
```

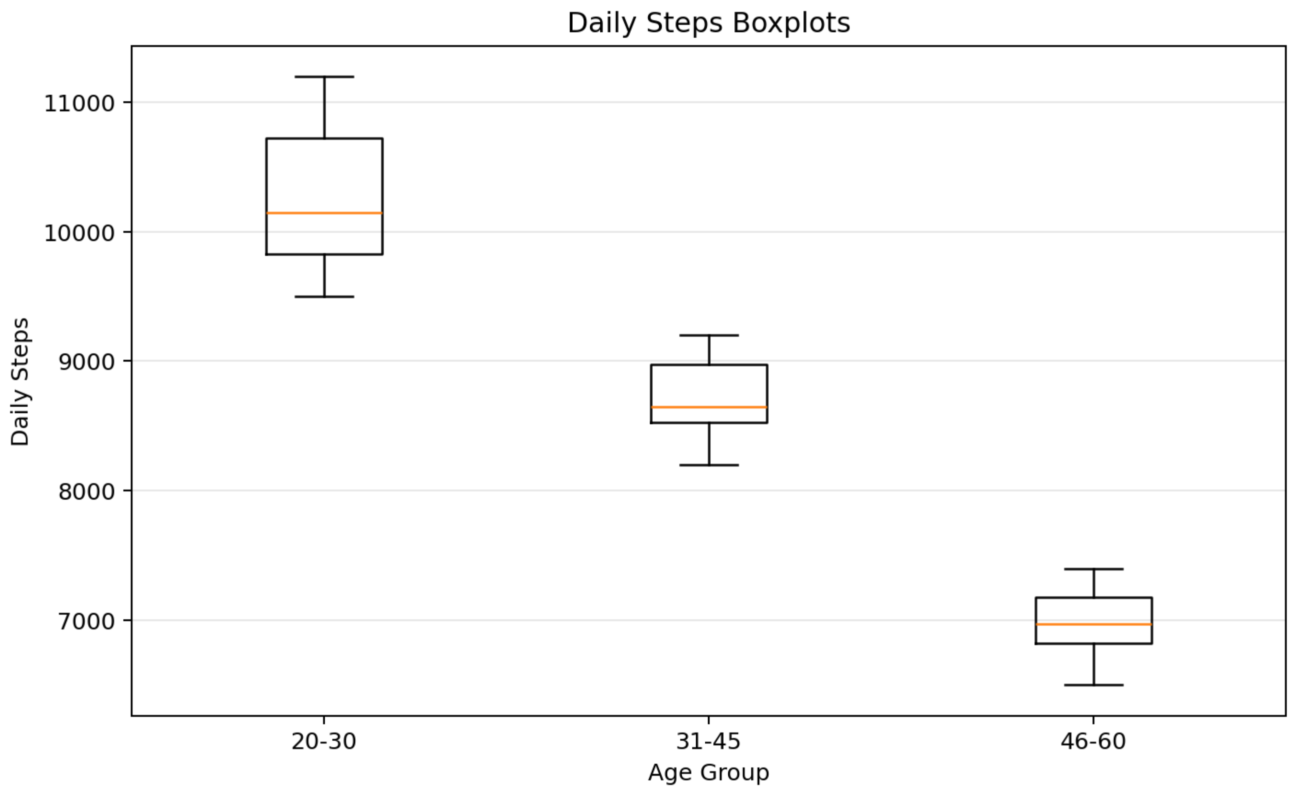
Output 1: Histograms



Output 2: Density Curves



Output 3: Boxplots



Questions and Answers

1. Draw histograms.

   Answer: The overlaid histograms above compare step-frequency distributions across the three age groups.

2. Plot density curves.

   Answer: The density curves above provide smooth comparisons of the daily-step distributions.

3. Create boxplots.

   Answer: The boxplots above compare medians, quartiles and overall spread among age groups.

4. Highest median group?

   Answer: The **20-30** age group has the highest median daily steps of **10,150**.

5. Best visualization and justify.

   Answer: The **boxplot** is the best single visualization for comparing these age groups because it clearly displays the median, spread and relative position of each distribution. Density plots are useful when distribution shape is the main focus.